Industry Sharing Session 3

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Outlines

Metaverse Platforms

- Persistent shared virtual spaces
- Focus on identity, social presence, digital ownership, and platform ecosystems
- Examples: virtual worlds, shared 3D spaces, metaverse-style environments

Spatial Installations

- Room-scale, embodied, multi-user experiences
- Designed through movement, proximity, sensory layering, and narrative
- Examples: projection spaces, interactive exhibitions, immersive rooms

AR Experiences

- Personal-device, browser-first interactions
- Fast access, guided onboarding, performance-aware design
- Examples: Web AR, mobile AR, lightweight interactive campaigns

What is Immersive Experience Design?

Not just 'cool tech' — it's a design philosophy rooted in how humans perceive, interact with, and emotionally respond to environments.

Presence

The psychological sense of 'being there.' Builds on Mel Slater's research: place illusion (feeling located in a virtual place) and plausibility illusion (events feel real). Tools: spatial audio, haptics, visual fidelity, responsive feedback loops.

Agency

The user isn't a passive viewer — they shape the experience through meaningful choices and interaction. This is what separates immersive design from cinema or traditional media. Key question: does the user feel their actions matter?

Emotion

Murray's concept of 'transformation' — the experience should change how you feel or think. Design for wonder, curiosity, connection. The emotional arc matters more than the polygon count.

Why This Matters

This isn't a niche — immersive experience design is becoming a core skill set across the design industry.

Expanding Design Canvases

Interaction design now includes AR overlays, projection-mapped rooms, and virtual environments. UX is going 3D.

Real Career Paths Opening Up

XR Designer, Spatial UX Researcher, Immersive Experience Producer — these roles didn't exist years ago. Companies like Niantic, Snap, and Meta are actively hiring.

New Interaction Paradigms

How do you design for gaze input? Body proximity? Shared physical-digital spaces? These require fundamentally different thinking from screen-based UX.

Critical Lens on Tech Hype

Metaverse, blockchain, AI — every new technology comes with inflated claims. Your training should help you separate design value from marketing noise.

The AR Hardware / Software Pipeline

Understanding the full stack helps you make better design decisions at every stage.

1

Concept & Design

Storyboarding, interaction mapping

→

2

3D Assets & Content

Modeling, animation, textures

→

3

AR Framework

8th Wall, ARKit, WebXR, Spark AR

→

4

Device Layer

Phone camera, LiDAR, sensors

→

5

User Experience

Real-time tracking, rendering, feedback

Interaction Design Across Scales

How does interaction design fundamentally change when the canvas shifts from a phone to a room?

Personal Device AR

- Touch, tap, swipe gestures
- Single-user, private viewport
- Small screen, limited FOV
- On-the-go, casual context
- Seconds to minutes sessions

Room-Scale Installations

- Body movement, proximity, gaze
- Multi-user, shared social space
- Surround visuals, full immersion
- Curated, intentional context
- Minutes to hours sessions

Design Insight: The fundamental shift isn't just input method — it's the relationship between user and environment. In personal AR, the user controls the frame. In spatial installations, the environment controls the experience.

Industry Trends: Where Is This Going?

WebAR

Browser-First AR Is the Default

8th Wall, WebXR, Safari AR — no-download experiences are growing 3x faster than app-based AR. Snap and Meta are investing billions in AR creator tools.

Spatial

Spatial Computing Goes Mainstream

Apple Vision Pro launched spatial computing for consumers. Meta Quest 3 sold millions. Designers who think in 3D space will define the next generation of interfaces.

AI + XR

Generative AI Meets Immersive

AI is accelerating 3D asset creation (Meshy, Luma), real-time environment generation, and personalized immersive experiences — lowering the barrier for designers.

$40B+

Enterprise & Retail Investment

IKEA Place, Sephora Virtual Artist, Nike AR try-ons, museum installations — immersive is no longer experimental, it's commercial strategy.

Blockchain & The Metaverse

Understanding the technology layer that enables digital ownership, identity, and persistent virtual worlds.

What is Blockchain?

A simplified mental model for designers — understanding the 'why' before the 'how'

Distributed Ledger

A shared record book that no single entity controls. Every participant holds a copy. Why it matters for design: no single point of failure, no gatekeeper.

Immutable Records

Once written, entries cannot be altered. This creates trust without a central authority. Why it matters: provenance and authenticity for digital creations.

Smart Contracts

Self-executing agreements coded into the chain. They automate rules and ownership transfers. Why it matters: creators can set their own terms.

Digital Ownership & Identity

How blockchain could enhance immersion through self-expression

- **Portable Identity**

  Your avatar, achievements, and digital assets follow you across virtual spaces.

- **Creator Ownership**

  Artists prove authorship and maintain rights — no platform intermediary needed.

- **Self-Expression**

  Unique digital items become part of how you present yourself in virtual worlds.

- **Interoperability**

  Assets created in one platform could potentially be used in another.

Try out this game!

What is The Metaverse?

The metaverse describes persistent, shared, 3D virtual spaces where people interact, create, and transact. It's not one product — it's an evolving concept built on real-time 3D, spatial computing, networking, and (potentially) blockchain. The term has been both over-promised and under-delivered.

What's Actually Real

- Spatial computing devices shipping (Vision Pro, Quest)
- Virtual events and concerts (Fortnite, Roblox)
- Enterprise training simulations
- Digital fashion and virtual try-on

What's Still Overhyped

- "One unified metaverse" replacing the internet
- Virtual real estate as investment
- Full interoperability between platforms
- Mass consumer adoption (still early)

How Blockchain Enhances Immersion

Provenance & Authenticity

Every digital creation has a verifiable origin story. In virtual galleries and immersive exhibitions, this adds meaning and trust.

Identity & Self-Expression

Blockchain-verified assets let users curate their virtual identity across platforms — your avatar, your creations, your history.

Persistent Economies

Virtual worlds can have real economies where creators are compensated and participants own their digital contributions.

Real-World Use Cases

Digital Art & Virtual Galleries

Artists use on-chain provenance to authenticate work in VR galleries. Feral File partners with leading digital artists; Art Blocks uses generative algorithms on-chain.

Wearable Digital Fashion

Nike (.SWOOSH), Adidas, Gucci are creating digital clothing wearable across virtual platforms. DressX enables virtual try-on with blockchain verification.

Token-Gated Immersive Events

Your ticket becomes a collectible that unlocks exclusive AR content or spatial experiences. Coachella, Art Basel have experimented with this model.

Collaborative Virtual Worlds

Decentraland and Spatial use blockchain for land ownership and asset persistence. Oncyber lets artists curate 3D gallery spaces with their owned works.

Connecting the Dots

Web AR → Design Principles

Measuring Immersion

How did the projects determine success? Engagement time, user behavior, emotional response — what signals indicate the experience 'worked'?

Design Constraints as Drivers

Battery drain, browser fragmentation, limited compute — these aren't just barriers. They shape creative decisions and force elegant solutions.

From Phone to Space

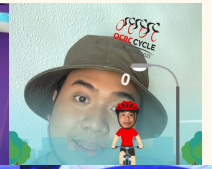
Next, we'll see how these same principles translate when the canvas expands from a 6-inch screen to an entire room.

Designing Web AR & Spatial Installation

Razi Afandi

2020

DIGIPEN DRAGONS



6 years

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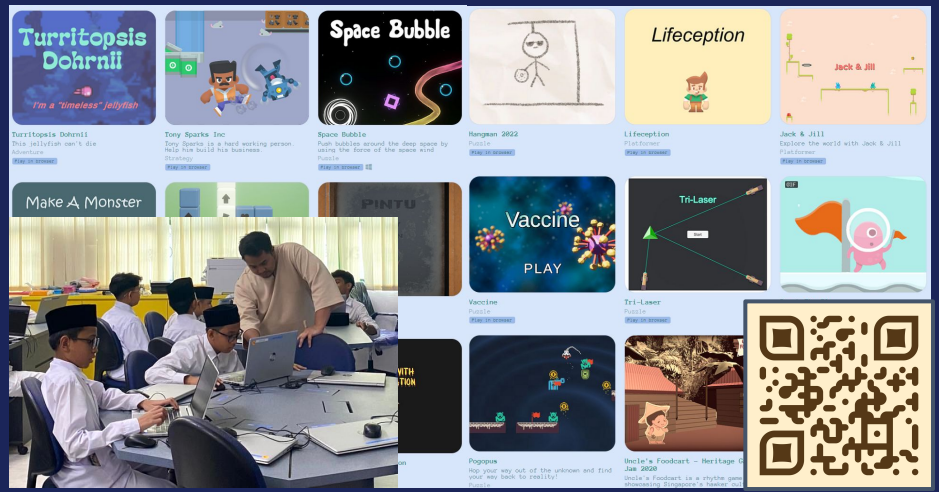

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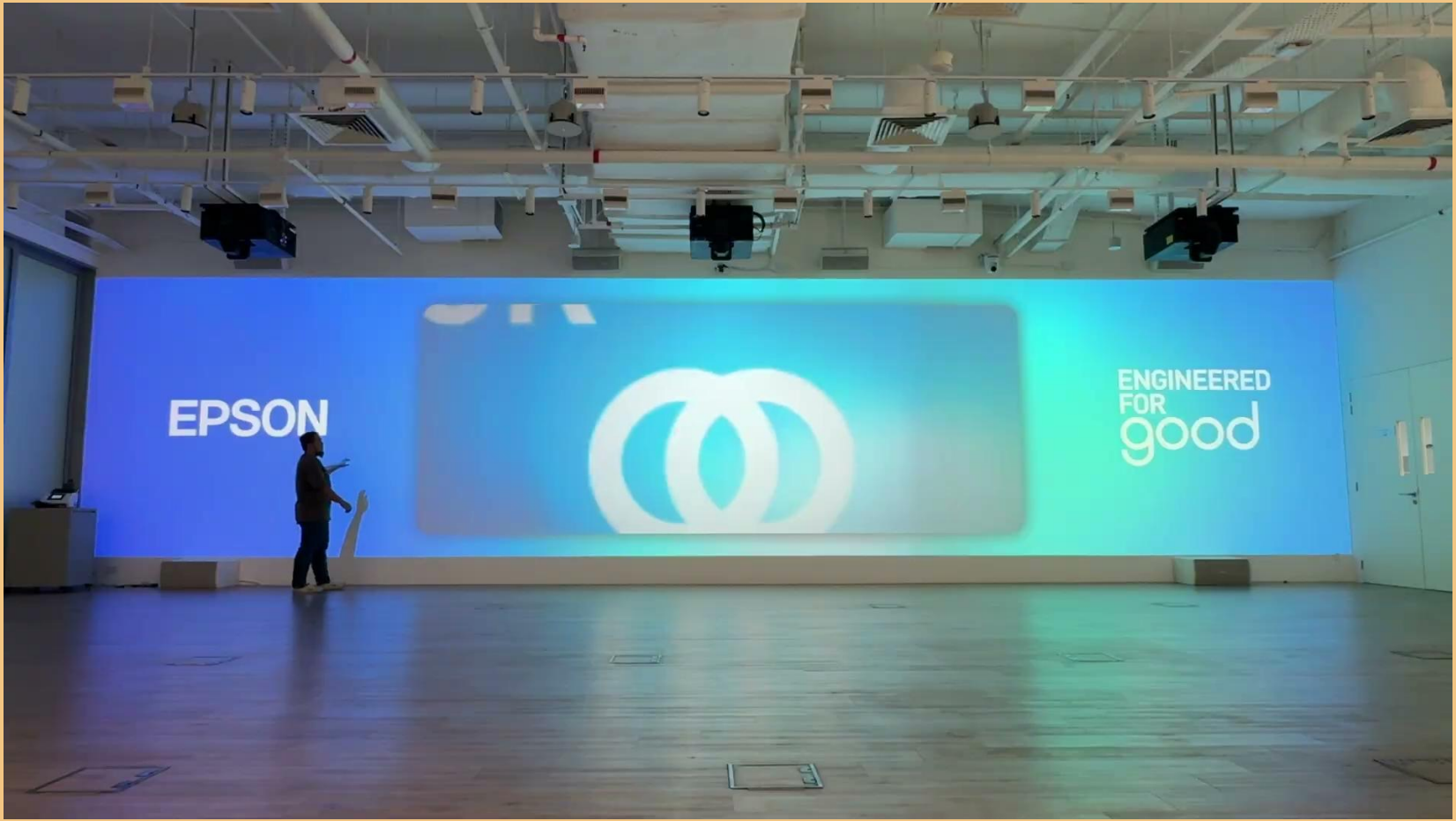
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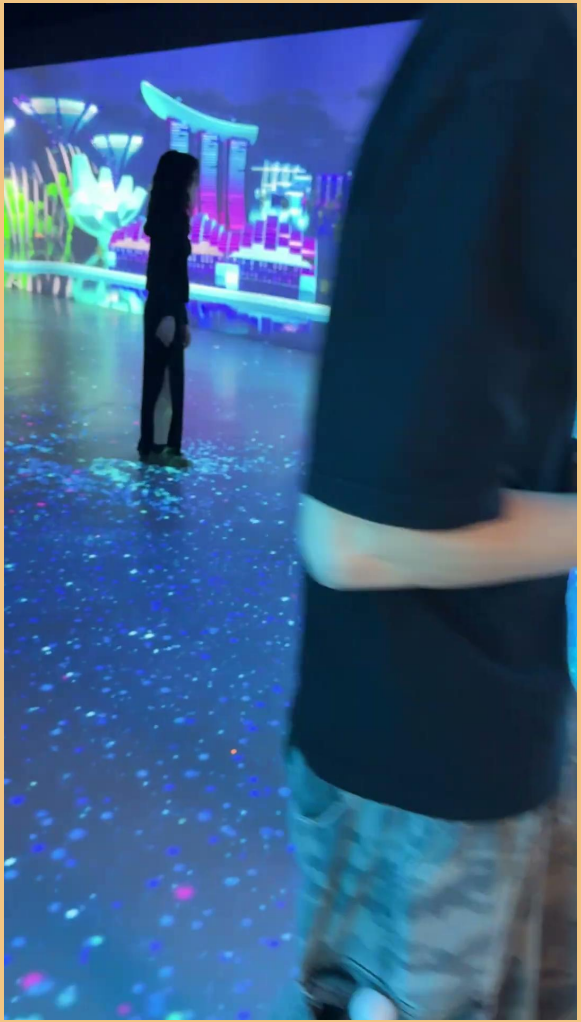

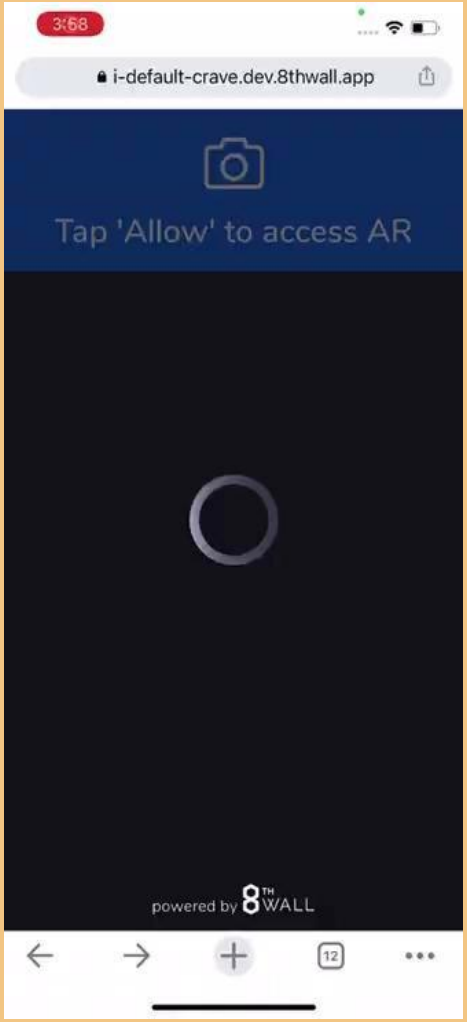
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Why Web AR?

The very first question before the commitment

Instant access	Browser-based	Frictionless	Tech Stack
Users simply scan a QR code or tap a link to launch	Runs directly in mobile browsers, no app download required	Eliminates the 70% user drop off rate associated with app installs	Powered by WebXR, 8th wall, AR.js and A-Frame using HTML/WebGL

Target Audience

Learn to read the room they said

Retail & E-commerce

Education / Training

Marketing

Art & Entertainment

Key Factors

These are the base foundations of making Web AR

Simplicity first

Minimize visual clutter, let content breathe

Context-Aware

Adapt to lighting and physical space

Guided Onboarding

Use visual cues and animations, not text

Performance

Optimize assets for smooth 60fps rendering

Barriers

There's always a way to overcome

Limited computing power: browser sandboxing constraints rendering & tracking capabilities

Battery Drain: high consumption from camera, sensors and real-time 3D rendering

Network Dependency: Latency and bandwidth limits cause lag; dependent on connection speed

Browser Fragmentation: Inconsistent webxr support across different browsers and devices

Strategies

OP meta strats

- **Asset Optimization**
  Use glTF/GLB, draco compression, and LOD systems
- **Progressive Loading**
  Stream 3D content in stages for instant feedback
- **Cloud Offloading**
  Shift heavy computation to cloud servers
- **Cross-Browser Testing**
  Ensure consistency across web browsers
- **Clear UX**
  Use visual cues to guide users

Key Factors

These are the base foundations of making Spatial Installations

Immersive Environments

The space is the canvas

Sensory Layers

projection mapping, LED Arrays, sensors and responsive audio

Active Participation

Audiences are not just viewers, but participants

Emotion First

Awe, curiosity, playfulness

Barriers

There's always a way to overcome

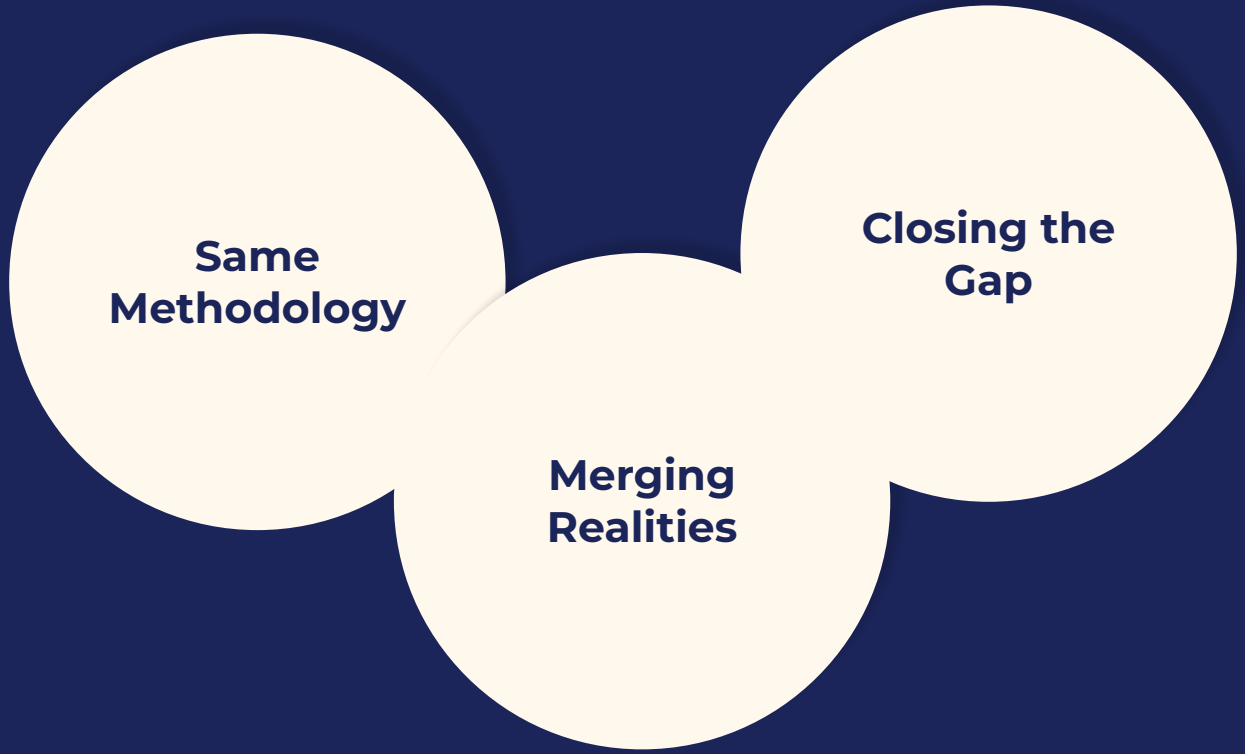
- **Environmental Constraints:** venue limits (lighting, acoustics, structure)
- **Technical Integration:** synching hardware (projectors, sensors) & software
- **Budget & Scale:** high costs for hardware, fabrications, and labor
- **Reliability:** need for auto-restart monitoring and robust design
- **Audience Unpredictability:** unexpected user interactions and safety

Strategies

OP meta strats

- **Narrative ARC**
  Map the visitor journey & emotional peeks
- **Sensory Layering**
  Combine light, sound, and video
- **Interaction Design**
  Define inputers (sensors) & outputs (visuals)
- **Accessibility**
  Design for all abilities & environments
- **Prototyping**
  Simulate & test before building

Conclusion



Now try this out!

Click me